

Appendix L

Air and $(CH_2)_n$ Properties at Low Pressure

Table L.1 Système International (SI) Units

<i>T</i>		<i>f</i> = 0		<i>f</i> = 0.0169		<i>f</i> = 0.0338		<i>f</i> = 0.0507		<i>f</i> = 0.0676	
K	°C	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>
170	−103	169.80	0.1903	170.82	0.1870	171.81	0.1840	172.76	0.1810	173.69	0.1783
180	−93	179.85	0.2324	180.95	0.2288	182.01	0.2254	183.04	0.2221	184.03	0.2190
190	−83	189.90	0.2809	191.08	0.2769	192.22	0.2731	193.33	0.2695	194.40	0.2661
200	−73	199.93	0.3361	201.21	0.3318	202.44	0.3278	203.63	0.3239	204.79	0.3202
210	−63	209.96	0.3985	211.34	0.3941	212.66	0.3899	213.95	0.3859	215.19	0.3821
220	−53	219.99	0.4688	221.47	0.4644	222.89	0.4602	224.28	0.4561	225.62	0.4522
230	−43	230.01	0.5475	231.60	0.5433	233.13	0.5392	234.62	0.5352	236.06	0.5315
240	−33	240.02	0.6352	241.73	0.6313	243.38	0.6276	244.98	0.6240	246.53	0.6205
250	−23	250.04	0.7325	251.87	0.7292	253.64	0.7260	255.36	0.7230	257.02	0.7201
260	−13	260.05	0.8399	262.01	0.8376	263.91	0.8353	265.75	0.8331	267.53	0.8310
270	−3	270.07	0.9581	272.17	0.9571	274.19	0.9561	276.16	0.9551	278.06	0.9542
280	7	280.08	1.0878	282.32	1.0885	284.49	1.0891	286.58	1.0898	288.61	1.0904
290	17	290.10	1.2296	292.49	1.2325	294.79	1.2352	297.03	1.2379	299.19	1.2406
300	27	300.13	1.3841	302.66	1.3897	305.11	1.3952	307.49	1.4005	309.79	1.4056
310	37	310.15	1.5521	312.85	1.5611	315.45	1.5699	317.97	1.5783	320.41	1.5865
320	47	320.19	1.7344	323.04	1.7474	325.80	1.7601	328.47	1.7724	331.05	1.7844
330	57	330.23	1.9315	333.25	1.9494	336.16	1.9667	338.99	1.9836	341.72	2.00
340	67	340.28	2.14	343.47	2.17	346.55	2.19	349.53	2.21	352.42	2.23
350	77	350.34	2.37	353.70	2.40	356.95	2.43	360.09	2.46	363.13	2.49
360	87	360.41	2.62	363.94	2.66	367.36	2.69	370.67	2.73	373.88	2.77

370	97	370.49	2.88	374.20	2.93	377.80	2.98	381.27	3.02	384.64	3.06
380	107	380.58	3.17	384.48	3.22	388.25	3.28	391.90	3.33	395.44	3.39
390	117	390.68	3.47	394.77	3.54	398.72	3.61	402.55	3.67	406.26	3.74
400	127	400.80	3.80	405.08	3.88	409.22	3.95	413.22	4.03	417.10	4.11
410	137	410.93	4.14	415.40	4.24	419.73	4.33	423.92	4.42	427.98	4.51
420	147	421.07	4.51	425.74	4.62	430.26	4.73	434.64	4.84	438.88	4.94
430	157	431.23	4.90	436.10	5.03	440.82	5.16	445.38	5.28	449.80	5.40
440	167	441.41	5.32	446.49	5.47	451.40	5.61	456.15	5.76	460.76	5.90
450	177	451.60	5.76	456.89	5.93	462.00	6.10	466.95	6.26	471.74	6.43
460	187	461.81	6.23	467.31	6.42	472.62	6.61	477.77	6.80	482.75	6.99
470	197	472.04	6.72	477.75	6.94	483.27	7.16	488.62	7.38	493.79	7.59
480	207	482.28	7.25	488.21	7.50	493.94	7.74	499.49	7.99	504.86	8.23
490	217	492.55	7.80	498.69	8.08	504.63	8.36	510.39	8.64	515.96	8.92
500	227	502.83	8.39	509.20	8.70	515.35	9.02	521.31	9.33	527.09	9.64
510	237	513.13	9.01	519.72	9.36	526.10	9.71	532.27	10.06	538.24	10.41
520	247	523.46	9.66	530.27	10.05	536.87	10.44	543.25	10.83	549.43	11.23
530	257	533.80	10.35	540.85	10.78	547.66	11.22	554.26	11.65	560.65	12.09
540	267	544.17	11.07	551.44	11.55	558.48	12.03	565.30	12.52	571.89	13.01
550	277	554.55	11.83	562.06	12.36	569.33	12.90	576.36	13.44	583.17	13.98
560	287	564.96	12.63	572.71	13.21	580.20	13.81	587.46	14.41	594.48	15.01
570	297	575.39	13.47	583.38	14.11	591.10	14.77	598.58	15.43	605.82	16.09
580	307	585.84	14.35	594.07	15.06	602.03	15.78	609.73	16.50	617.19	17.24

(Continued)

Table L.1 Système International (SI) Units *Continued*)

<i>T</i>		<i>f</i> = 0		<i>f</i> = 0.0169		<i>f</i> = 0.0338		<i>f</i> = 0.0507		<i>f</i> = 0.0676	
K	°C	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>
590	317	596.32	15.27	604.79	16.05	612.98	16.84	620.91	17.64	628.59	18.45
600	327	606.81	16.24	615.53	17.09	623.96	17.96	632.12	18.83	640.02	19.72
600	327	606.81	16.24	615.53	17.09	623.96	17.96	632.12	18.83	640.02	19.72
610	337	617.33	17.25	626.30	18.18	634.97	19.13	643.36	20.09	651.49	21.06
620	347	627.88	18.32	637.09	19.33	646.00	20.36	654.63	21.41	662.98	22.48
630	357	638.44	19.43	647.91	20.53	657.07	21.66	665.93	22.80	674.51	23.97
640	367	649.03	20.59	658.75	21.79	668.16	23.01	677.25	24.26	686.07	25.53
650	377	659.65	21.80	669.62	23.11	679.27	24.44	688.61	25.79	697.65	27.18
660	387	670.29	23.07	680.52	24.48	690.42	25.93	700.00	27.40	709.28	28.91
670	397	680.95	24.40	691.44	25.93	701.59	27.49	711.42	29.09	720.93	30.72
680	407	691.63	25.78	702.39	27.43	712.79	29.12	722.86	30.86	732.61	32.63
690	417	702.34	27.23	713.36	29.01	724.02	30.83	734.34	32.71	744.33	34.63
700	427	713.07	28.73	724.36	30.65	735.28	32.62	745.84	34.65	756.08	36.72
710	437	723.83	30.30	735.38	32.36	746.56	34.49	757.38	36.68	767.85	38.92
720	447	734.61	31.93	746.43	34.15	757.87	36.44	768.94	38.80	779.67	41.22
730	457	745.41	33.64	757.51	36.02	769.21	38.48	780.54	41.02	791.51	43.63
740	467	756.24	35.41	768.61	37.97	780.58	40.61	792.16	43.34	803.38	46.15
750	477	767.09	37.25	779.74	39.99	791.98	42.83	803.82	45.76	815.29	48.79
760	487	777.97	39.17	790.90	42.10	803.40	45.15	815.50	48.29	827.22	51.54

770	497	788.87	41.16	802.08	44.30	814.85	47.56	827.22	50.94	839.19	54.42
780	507	799.79	43.23	813.28	46.59	826.33	50.08	838.96	53.69	851.19	57.43
790	517	810.74	45.39	824.51	48.97	837.83	52.70	850.73	56.57	863.21	60.58
800	527	821.71	47.62	835.77	51.45	849.37	55.43	862.53	59.57	875.27	63.86
810	537	832.70	49.94	847.05	54.02	860.93	58.27	874.36	62.69	887.36	67.28
820	547	843.72	52.35	858.35	56.69	872.51	61.22	886.22	65.94	899.48	70.85
830	557	854.76	54.85	869.69	59.47	884.13	64.30	898.10	69.33	911.64	74.57
840	567	865.82	57.44	881.04	62.36	895.77	67.50	910.02	72.86	923.82	78.45
850	577	876.90	60.13	892.42	65.35	907.43	70.82	921.96	76.53	936.03	82.49
860	587	888.01	62.91	903.83	68.46	919.12	74.27	933.93	80.35	948.27	86.70
870	597	899.14	65.80	915.26	71.68	930.84	77.86	945.93	84.33	960.54	91.09
880	607	910.29	68.79	926.71	75.02	942.59	81.58	957.96	88.46	972.84	95.65
890	617	921.47	71.88	938.19	78.49	954.36	85.44	970.01	92.75	985.17	100.40
900	627	932.66	75.08	949.69	82.08	966.15	89.46	982.09	97.21	997.52	105.33
910	637	943.88	78.40	961.21	85.80	977.97	93.62	994.20	101.84	1009.91	110.46
920	647	955.12	81.82	972.76	89.66	989.82	97.93	1006.33	106.64	1022.32	115.80
930	657	966.38	85.37	984.33	93.65	1001.69	102.40	1018.49	111.64	1034.77	121.34
940	667	977.66	89.03	995.92	97.78	1013.58	107.04	1030.68	116.81	1047.24	127.10
950	677	988.96	92.82	1007.54	102.06	1025.50	111.84	1042.89	122.19	1059.73	133.09
960	687	1000.28	96.74	1019.17	106.48	1037.45	116.82	1055.13	127.76	1072.26	139.29
970	697	1011.62	100.78	1030.83	111.06	1049.41	121.97	1067.40	133.53	1084.81	145.74
980	707	1022.99	104.96	1042.51	115.79	1061.41	127.31	1079.69	139.52	1097.39	152.43

(Continued)

Table L.1 Système International (SI) Units *Continued*)

<i>T</i>		<i>f</i> = 0		<i>f</i> = 0.0169		<i>f</i> = 0.0338		<i>f</i> = 0.0507		<i>f</i> = 0.0676	
K	°C	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>
990	717	1034.37	109.27	1054.22	120.68	1073.42	132.83	1092.00	145.72	1110.00	159.36
1000	727	1045.77	113.72	1065.94	125.74	1085.46	138.54	1104.34	152.14	1122.63	166.55
1000	727	1045.77	113.72	1065.94	125.74	1085.46	138.54	1104.34	152.14	1122.63	166.55
1010	737	1057.19	118.31	1077.69	130.96	1097.52	144.45	1116.71	158.79	1135.29	174.01
1020	747	1068.63	123.05	1089.45	136.36	1109.60	150.56	1129.09	165.68	1147.97	181.74
1030	757	1080.09	127.94	1101.24	141.93	1121.70	156.88	1141.51	172.81	1160.68	189.74
1040	767	1091.56	132.98	1113.05	147.68	1133.83	163.41	1153.94	180.19	1173.42	198.03
1050	777	1103.06	138.18	1124.87	153.62	1145.98	170.15	1166.40	187.82	1186.18	206.6
1060	787	1114.57	143.53	1136.72	159.74	1158.15	177.13	1178.89	195.71	1198.97	215.5
1070	797	1126.10	149.05	1148.59	166.06	1170.34	184.32	1191.39	203.9	1211.78	224.7
1080	807	1137.65	154.73	1160.47	172.58	1182.55	191.76	1203.92	212.3	1224.61	234.2
1090	817	1149.22	160.59	1172.38	179.30	1194.78	199.43	1216.47	221.0	1237.47	244.1
1100	827	1160.80	166.62	1184.30	186.23	1207.04	207.4	1229.04	230.0	1250.35	254.3
1110	837	1172.40	172.83	1196.24	193.38	1219.31	215.5	1241.64	239.3	1263.26	264.8
1120	847	1184.01	179.22	1208.21	200.7	1231.61	224.0	1254.25	248.9	1276.18	275.7
1130	857	1195.65	185.79	1220.18	208.3	1243.92	232.7	1266.89	258.9	1289.13	287.0
1140	867	1207.29	192.55	1232.18	216.1	1256.25	241.6	1279.55	269.1	1302.11	298.6
1150	877	1218.96	199.51	1244.19	224.2	1268.60	250.9	1292.23	279.7	1315.10	310.6
1160	887	1230.64	206.7	1256.22	232.4	1280.97	260.4	1304.93	290.6	1328.12	323.0

1170	897	1242.33	214.0	1268.27	241.0	1293.36	270.2	1317.65	301.8	1341.16	335.9
1180	907	1254.05	221.6	1280.34	249.8	1305.77	280.3	1330.39	313.4	1354.22	349.1
1190	917	1265.77	229.4	1292.42	258.8	1318.20	290.8	1343.15	325.4	1367.30	362.8
1200	927	1277.51	237.4	1304.52	268.1	1330.64	301.5	1355.92	337.7	1380.41	376.8
1210	937	1289.27	245.6	1316.63	277.6	1343.10	312.5	1368.72	350.4	1393.53	391.4
1220	947	1301.03	254.0	1328.76	287.4	1355.58	323.9	1381.54	363.5	1406.68	406.4
1230	957	1312.82	262.6	1340.91	297.5	1368.08	335.6	1394.38	377.0	1419.84	421.9
1240	967	1324.61	271.5	1353.07	307.9	1380.59	347.7	1407.23	390.9	1433.02	437.8
1250	977	1336.43	280.6	1365.24	318.6	1393.12	360.1	1420.10	405.2	1446.23	454.3
1260	987	1348.25	290.0	1377.44	329.6	1405.67	372.8	1432.99	420.0	1459.45	471.2
1270	997	1360.09	299.6	1389.64	340.8	1418.23	385.9	1445.90	435.2	1472.69	488.7
1280	1007	1371.94	309.5	1401.86	352.4	1430.81	399.4	1458.83	450.8	1485.96	506.7
1290	1017	1383.80	319.6	1414.10	364.3	1443.41	413.3	1471.77	466.8	1499.24	525.2
1300	1027	1395.68	330.0	1426.35	376.5	1456.02	427.5	1484.73	483.4	1512.54	544.3
1310	1037	1407.57	340.6	1438.61	389.0	1468.64	442.2	1497.71	500.4	1525.85	564.0
1320	1047	1419.47	351.6	1450.89	401.8	1481.29	457.2	1510.70	517.9	1539.19	584.2
1330	1057	1431.38	362.7	1463.18	415.0	1493.94	472.6	1523.71	535.9	1552.54	605.0
1340	1067	1443.31	374.2	1475.49	428.6	1506.61	488.5	1536.74	554.4	1565.91	626.5
1350	1077	1455.25	386.0	1487.81	442.5	1519.30	504.8	1549.78	573.4	1579.30	648.5
1360	1087	1467.20	398.0	1500.14	456.7	1532.00	521.6	1562.84	593.0	1592.70	671.2
1370	1097	1479.16	410.3	1512.48	471.3	1544.72	538.8	1575.92	613.1	1606.13	694.6
1380	1107	1491.13	423.0	1524.84	486.3	1557.45	556.4	1589.01	633.7	1619.57	718.6

(Continued)

Table L.1 Système International (SI) Units *Continued*)

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K	°C	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>
1390	1117	1503.11	435.9	1537.21	501.7	1570.19	574.5	1602.11	654.9	1633.02	743.3
1400	1127	1515.11	449.2	1549.59	517.4	1582.95	593.1	1615.23	676.7	1646.49	768.6
1400	1127	1515.11	449.2	1549.59	517.4	1582.95	593.1	1615.23	676.7	1646.49	768.6
1410	1137	1527.11	462.8	1561.99	533.6	1595.72	612.2	1628.37	699.1	1659.98	794.7
1420	1147	1539.13	476.7	1574.39	550.1	1608.51	631.7	1641.52	722.0	1673.49	821.5
1430	1157	1551.16	490.9	1586.81	567.1	1621.30	651.8	1654.68	745.6	1687.01	849.1
1440	1167	1563.20	505.4	1599.25	584.4	1634.12	672.4	1667.86	769.8	1700.54	877.4
1450	1177	1575.25	520.4	1611.69	602.2	1646.94	693.4	1681.06	794.6	1714.09	906.5
1460	1187	1587.31	535.6	1624.14	620.4	1659.78	715.1	1694.27	820.2	1727.66	936.4
1470	1197	1599.38	551.2	1636.61	639.1	1672.63	737.2	1707.49	846.3	1741.24	967.0
1480	1207	1611.46	567.2	1649.09	658.2	1685.49	760.0	1720.72	873.1	1754.84	998.5
1490	1217	1623.55	583.5	1661.58	677.8	1698.37	783.2	1733.97	900.7	1768.45	1030.8
1500	1227	1635.65	600.2	1674.08	697.8	1711.26	807.1	1747.24	928.9	1782.08	1064.0
1510	1237	1647.76	617.2	1686.59	718.3	1724.16	831.5	1760.51	957.8	1795.72	1098.1
1520	1247	1659.88	634.7	1699.11	739.3	1737.07	856.6	1773.80	987.5	1809.38	1133.1
1530	1257	1672.01	652.5	1711.65	760.7	1749.99	882.2	1787.11	1018.0	1823.05	1168.9
1540	1267	1684.15	670.7	1724.19	782.7	1762.93	908.5	1800.42	1049.1	1836.73	1205.7
1550	1277	1696.29	689.4	1736.75	805.2	1775.88	935.4	1813.75	1081.1	1850.43	1243.5
1560	1287	1708.45	708.4	1749.31	828.2	1788.84	962.9	1827.10	1113.9	1864.14	1282.2

1570	1297	1720.62	727.9	1761.89	851.6	1801.81	991.1	1840.45	1147.4	1877.86	1321.9
1580	1307	1732.80	747.7	1774.48	875.7	1814.80	1019.9	1853.82	1181.8	1891.60	1362.6
1590	1317	1744.98	768.0	1787.08	900.3	1827.79	1049.5	1867.20	1217.0	1905.36	1404.3
1600	1327	1757.18	788.8	1799.68	925.4	1840.80	1079.7	1880.59	1253.1	1919.12	1447.1
1610	1337	1769.38	810.0	1812.30	951.1	1853.81	1110.6	1893.99	1290.0	1932.90	1490.9
1620	1347	1781.59	831.6	1824.93	977.3	1866.84	1142.2	1907.41	1327.8	1946.69	1535.9
1630	1357	1793.82	853.7	1837.56	1004.2	1879.88	1174.6	1920.84	1366.6	1960.50	1581.9
1640	1367	1806.05	876.2	1850.21	1031.6	1892.93	1207.6	1934.28	1406.2	1974.31	1629.1
1650	1377	1818.29	899.2	1862.87	1059.6	1905.99	1241.5	1947.73	1446.8	1988.15	1677.4
1660	1387	1830.54	922.7	1875.54	1088.2	1919.06	1276.1	1961.19	1488.3	2001.99	1726.9
1670	1397	1842.79	946.7	1888.21	1117.4	1932.15	1311.5	1974.67	1530.8	2015.84	1777.6
1680	1407	1855.06	971.2	1900.90	1147.3	1945.24	1347.6	1988.15	1574.3	2029.71	1829.5
1690	1417	1867.33	996.1	1913.59	1177.8	1958.34	1384.6	2001.65	1618.8	2043.59	1882.6
1700	1427	1879.61	1021.6	1926.30	1209.0	1971.45	1422.4	2015.16	1664.3	2057.48	1937.0
1710	1437	1891.90	1047.6	1939.01	1240.8	1984.58	1461.0	2028.68	1710.8	2071.38	1992.8
1720	1447	1904.20	1074.1	1951.73	1273.2	1997.71	1500.5	2042.21	1758.4	2085.30	2049.8
1730	1457	1916.51	1101.1	1964.46	1306.4	2010.85	1540.8	2055.75	1807.1	2099.23	2108.1
1740	1467	1928.82	1128.7	1977.20	1340.2	2024.01	1582.0	2069.30	1856.9	2113.16	2167.8
1750	1477	1941.14	1156.8	1989.95	1374.8	2037.17	1624.1	2082.86	1907.8	2127.11	2229
1760	1487	1953.47	1185.5	2002.71	1410.0	2050.34	1667.1	2096.44	1959.8	2141.07	2291
1770	1497	1965.81	1214.7	2015.48	1446.0	2063.52	1711.0	2110.02	2013.0	2155.05	2355
1780	1507	1978.16	1244.5	2028.25	1482.7	2076.71	1755.8	2123.61	2067.3	2169.03	2421
1790	1517	1990.51	1274.9	2041.04	1520.1	2089.91	1801.6	2137.22	2122.8	2183.02	2488

(Continued)

Table L.1 Système International (SI) Units *Continued*)

<i>T</i>		<i>f</i> = 0		<i>f</i> = 0.0169		<i>f</i> = 0.0338		<i>f</i> = 0.0507		<i>f</i> = 0.0676	
K	°C	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>	<i>h</i> , kJ/kg	<i>P_r</i>
1800	1527	2002.87	1305.8	2053.83	1558.3	2103.12	1848.3	2150.83	2179.6	2197.03	2556
1810	1537	2015.24	1337.4	2066.63	1597.3	2116.34	1896.0	2164.45	2238	2211.04	2626
1820	1547	2027.61	1369.5	2079.44	1637.0	2129.57	1944.8	2178.08	2297	2225.07	2697
1830	1557	2040.00	1402.3	2092.25	1677.6	2142.80	1994.5	2191.73	2357	2239.10	2771
1840	1567	2052.39	1435.7	2105.08	1718.9	2156.05	2045.2	2205.38	2419	2253.15	2845
1850	1577	2064.78	1469.7	2117.91	1761.0	2169.30	2097	2219.04	2482	2267.20	2922
1860	1587	2077.19	1504.3	2130.75	1804.0	2182.56	2150	2232.71	2547	2281.27	3000
1870	1597	2089.60	1539.6	2143.60	1847.8	2195.83	2204	2246.39	2613	2295.34	3080
1880	1607	2102.02	1575.6	2156.45	1892.4	2209.11	2259	2260.08	2680	2309.43	3161
1890	1617	2114.44	1612.2	2169.32	1938.0	2222.40	2315	2273.77	2748	2323.52	3244
1900	1627	2126.87	1649.4	2182.19	1984.3	2235.69	2372	2287.48	2818	2337.63	3329
1910	1637	2139.31	1687.4	2195.06	2031.6	2249.00	2430	2301.19	2890	2351.74	3416
1920	1647	2151.75	1726.0	2207.95	2079.8	2262.31	2490	2314.92	2963	2365.86	3505
1930	1657	2164.20	1765.4	2220.84	2128.8	2275.62	2551	2328.65	3037	2379.99	3595
1940	1667	2176.66	1805.4	2233.74	2178.8	2288.95	2612	2342.39	3113	2394.13	3688
1950	1677	2189.12	1846.2	2246.64	2230	2302.28	2676	2356.13	3191	2408.28	3782
1960	1687	2201.59	1887.7	2259.55	2282	2315.62	2740	2369.89	3270	2422.43	3879
1970	1697	2214.06	1929.9	2272.47	2334	2328.97	2805	2383.65	3350	2436.60	3977
1980	1707	2226.54	1972.8	2285.40	2388	2342.32	2872	2397.42	3433	2450.77	4078

1990	1717	2239.03	2016.6	2298.33	2443	2355.68	2940	2411.20	3516	2464.95	4180
2000	1727	2251.52	2061	2311.26	2499	2369.05	3010	2424.98	3602	2479.14	4285
2010	1737	2264.02	2106	2324.21	2556	2382.43	3080	2438.77	3689	2493.34	4392
2020	1747	2276.52	2152	2337.16	2614	2395.81	3152	2452.57	3778	2507.54	4501
2030	1757	2289.03	2199	2350.11	2672	2409.20	3226	2466.38	3869	2521.75	4612
2040	1767	2301.55	2247	2363.07	2732	2422.59	3301	2480.19	3961	2535.97	4725
2050	1777	2314.07	2295	2376.04	2793	2435.99	3377	2494.01	4055	2550.20	4841
2060	1787	2326.59	2345	2389.01	2855	2449.40	3454	2507.84	4152	2564.43	4959
2070	1797	2339.12	2395	2401.99	2919	2462.81	3533	2521.67	4249	2578.67	5079
2080	1807	2351.65	2446	2414.98	2983	2476.23	3614	2535.52	4349	2592.92	5202
2090	1817	2364.19	2497	2427.97	3048	2489.66	3696	2549.36	4451	2607.18	5327
2100	1827	2376.74	2550	2440.96	3115	2503.09	3779	2563.22	4554	2621.44	5455
2110	1837	2389.29	2604	2453.97	3183	2516.53	3864	2577.07	4660	2635.71	5585
2120	1847	2401.85	2658	2466.97	3252	2529.97	3950	2590.94	4767	2649.98	5717
2130	1857	2414.41	2713	2479.98	3322	2543.42	4038	2604.81	4877	2664.26	5853
2140	1867	2426.97	2769	2493.00	3393	2556.88	4128	2618.69	4988	2678.55	5990
2150	1877	2439.54	2827	2506.03	3465	2570.34	4219	2632.58	5102	2692.85	6131
2160	1887	2452.12	2885	2519.06	3539	2583.80	4312	2646.47	5218	2707.15	6274
2170	1897	2464.70	2944	2532.09	3614	2597.28	4406	2660.37	5336	2721.46	6420
2180	1907	2477.29	3004	2545.13	3690	2610.76	4502	2674.27	5456	2735.77	6568
2190	1917	2489.88	3065	2558.18	3768	2624.24	4600	2688.18	5578	2750.10	6720
2200	1927	2502.48	3126	2571.23	3847	2637.73	4699	2702.10	5702	2764.43	6874
2210	1937	2515.08	3189	2584.29	3927	2651.23	4801	2716.02	5829	2778.76	7031
2220	1947	2527.69	3253	2597.35	4008	2664.74	4904	2729.95	5958	2793.11	7191

Table L.2 British Engineering (BE) Units

<i>T</i>		<i>f</i> = 0		<i>f</i> = 0.0169		<i>f</i> = 0.0338		<i>f</i> = 0.0507		<i>f</i> = 0.0676	
°R	°F	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>
300	−160	71.55	0.1775	71.98	0.1744	72.40	0.1715	72.80	0.1687	73.19	0.1660
320	−140	76.36	0.2225	76.82	0.2190	77.27	0.2156	77.71	0.2124	78.13	0.2094
340	−120	81.16	0.2752	81.66	0.2712	82.15	0.2675	82.62	0.2639	83.08	0.2605
360	−100	85.95	0.3361	86.50	0.3318	87.03	0.3278	87.54	0.3239	88.04	0.3202
380	−80	90.74	0.4059	91.34	0.4015	91.91	0.3973	92.47	0.3933	93.01	0.3894
400	−60	95.53	0.4856	96.18	0.4812	96.80	0.4769	97.40	0.4729	97.99	0.4690
420	−40	100.32	0.5757	101.02	0.5716	101.69	0.5675	102.35	0.5637	102.98	0.5600
440	−20	105.10	0.6772	105.86	0.6736	106.59	0.6700	107.30	0.6667	107.99	0.6634
460	0	109.88	0.7909	110.70	0.7881	111.49	0.7854	112.26	0.7828	113.00	0.7803
480	20	114.67	0.9175	115.55	0.9160	116.40	0.9145	117.23	0.9131	118.03	0.9117
500	40	119.45	1.0580	120.40	1.0582	121.32	1.0585	122.21	1.0587	123.07	1.0589
520	60	124.24	1.213	125.25	1.216	126.24	1.218	127.19	1.221	128.12	1.223
540	80	129.02	1.384	130.11	1.390	131.17	1.395	132.19	1.400	133.18	1.406
560	100	133.81	1.572	134.98	1.581	136.11	1.590	137.20	1.599	138.25	1.608
580	120	138.61	1.777	139.85	1.791	141.05	1.805	142.21	1.818	143.34	1.831
600	140	143.41	2.00	144.73	2.02	146.00	2.04	147.24	2.06	148.44	2.08
620	160	148.21	2.24	149.61	2.27	150.97	2.30	152.28	2.32	153.55	2.35
640	180	153.01	2.51	154.50	2.54	155.94	2.58	159.33	2.61	158.68	2.64
660	200	157.83	2.79	159.40	2.84	160.92	2.88	162.39	2.92	163.81	2.96

680	220	162.65	3.10	164.30	3.16	165.91	3.21	167.46	3.26	168.97	3.31
700	240	167.47	3.44	169.22	3.50	170.91	3.57	172.55	3.63	174.13	3.70
720	260	172.30	3.80	174.14	3.88	175.92	3.95	177.64	4.03	179.31	4.11
740	280	177.14	4.18	179.07	4.28	180.94	4.37	182.75	4.47	184.51	4.56
760	300	181.99	4.59	184.02	4.71	185.98	4.82	187.88	4.93	189.72	5.04
780	320	186.84	5.04	188.97	5.17	191.02	5.30	193.01	5.44	194.94	5.57
800	340	191.71	5.51	193.93	5.67	196.08	5.82	198.16	5.98	200.18	6.13
820	360	196.58	6.02	198.90	6.20	201.15	6.38	203.32	6.56	205.43	6.74
840	380	201.46	6.56	203.89	6.77	206.23	6.97	208.50	7.18	210.70	7.39
860	400	206.35	7.13	208.88	7.37	211.32	7.61	213.69	7.85	215.98	8.09
880	420	211.25	7.74	213.89	8.02	216.43	8.29	218.89	8.57	221.28	8.84
900	440	216.17	8.39	218.90	8.70	221.55	9.02	224.11	9.33	226.59	9.64
920	460	221.09	9.08	223.93	9.43	226.68	9.79	229.35	10.14	231.92	10.50
940	480	226.02	9.81	228.97	10.21	231.83	10.61	234.59	11.01	237.27	11.42
960	500	230.96	10.58	234.03	11.03	236.99	11.48	239.86	11.94	242.63	12.39
980	520	235.92	11.40	239.09	11.91	242.16	12.41	245.13	12.92	248.01	13.44
1000	540	240.89	12.27	244.17	12.83	247.35	13.40	250.43	13.97	253.40	14.54
1020	560	245.86	13.18	249.26	13.81	252.55	14.44	255.73	15.08	258.82	15.72
1040	580	250.85	14.15	254.37	14.84	257.77	15.55	261.06	16.26	264.24	16.98
1060	600	255.86	15.17	259.49	15.94	263.00	16.72	266.39	17.51	269.69	18.31
1080	620	260.87	16.24	264.62	17.09	268.24	17.96	271.75	18.83	275.15	19.72
1100	640	265.89	17.37	269.76	18.31	273.50	19.26	277.12	20.23	280.62	21.22

(Continued)

Table L.2 British Engineering (BE) Units *Continued*

<i>T</i>		<i>f</i> = 0		<i>f</i> = 0.0169		<i>f</i> = 0.0338		<i>f</i> = 0.0507		<i>f</i> = 0.0676	
°R	°F	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>
1120	660	270.93	18.56	274.92	19.59	278.77	20.64	282.50	21.72	286.12	22.80
1140	680	275.98	19.81	280.09	20.94	284.06	22.10	287.90	23.28	291.63	24.48
1160	700	281.05	21.12	285.27	22.37	289.36	23.64	293.32	24.93	297.15	26.25
1180	720	286.12	22.50	290.47	23.86	294.68	25.26	298.75	26.68	302.70	28.13
1200	740	291.21	23.95	295.68	25.44	300.01	26.96	304.20	28.52	308.26	30.11
1220	760	296.31	25.47	300.91	27.09	305.36	28.75	309.66	30.46	313.83	32.20
1240	780	301.42	27.06	306.15	28.83	310.72	30.64	315.14	32.50	319.43	34.40
1260	800	306.55	28.73	311.40	30.65	316.10	32.62	320.64	34.65	325.04	36.72
1280	820	311.69	30.48	316.67	32.56	321.49	34.70	326.15	36.91	330.66	39.17
1300	840	316.84	32.31	321.95	34.56	326.89	36.89	331.68	39.28	336.31	41.75
1320	860	322.00	34.22	327.24	36.66	332.31	39.18	337.22	41.78	341.97	44.46
1340	880	327.18	36.22	332.55	38.86	337.75	41.59	342.78	44.40	347.65	47.31
1360	900	332.37	38.28	337.87	41.16	343.20	44.11	348.35	47.16	353.34	50.30
1380	920	337.57	40.49	343.21	43.64	348.66	46.95	353.94	50.39	359.05	53.95
1400	940	342.79	42.77	348.56	46.07	354.14	49.51	359.54	53.07	364.78	56.75
1420	960	348.01	45.14	353.92	48.70	359.63	52.40	365.17	56.24	370.52	60.22
1440	980	353.25	47.62	359.30	51.45	365.14	55.43	370.80	59.57	376.28	63.86
1460	1000	358.50	50.21	364.69	54.31	370.67	58.59	376.45	63.04	382.06	67.67
1480	1020	363.77	52.90	370.09	57.30	376.20	61.90	382.12	66.69	387.85	71.66

1500	1040	369.04	55.70	375.50	60.42	381.75	65.35	387.80	70.49	393.66	75.85
1520	1060	374.33	58.62	380.93	63.67	387.32	68.96	393.50	74.48	399.48	80.23
1540	1080	379.63	61.66	386.37	67.06	392.90	72.72	399.21	78.64	405.32	84.81
1560	1100	384.95	64.83	391.83	70.59	398.49	76.65	404.93	82.98	411.18	89.61
1580	1120	390.27	68.11	397.30	74.27	404.10	80.74	410.68	87.52	417.05	94.62
1600	1140	395.60	71.53	402.78	78.10	409.72	85.01	416.43	92.26	422.93	99.86
1620	1160	400.95	75.08	408.27	82.08	415.35	89.46	422.20	97.21	428.83	105.33
1640	1180	406.31	78.77	413.78	86.22	421.00	94.09	427.98	102.36	434.75	111.05
1660	1200	411.68	82.60	419.29	90.53	426.66	98.91	433.78	107.74	440.68	117.01
1680	1220	417.06	86.58	424.82	95.01	432.33	103.93	439.60	113.34	446.63	123.24
1700	1240	422.45	90.70	430.36	99.66	438.02	109.15	445.42	119.18	452.59	129.73
1720	1260	427.86	94.98	435.92	104.50	443.72	114.59	451.26	125.26	458.57	136.51
1740	1280	433.27	99.42	441.48	109.52	449.43	120.23	457.12	131.58	464.56	143.56
1760	1300	438.70	104.02	447.06	114.73	455.15	126.11	462.98	138.17	470.57	150.92
1780	1320	444.13	108.78	452.65	120.13	460.89	132.21	468.86	145.02	476.59	158.58
1800	1340	449.58	113.72	458.25	125.74	466.64	138.54	474.76	152.14	482.62	166.55
1820	1360	455.03	118.83	463.86	131.55	472.40	145.12	480.66	159.55	488.67	174.85
1840	1380	460.50	124.13	469.48	137.58	478.17	151.95	486.58	167.24	494.73	183.49
1860	1400	465.97	129.60	475.11	143.82	483.96	159.03	492.51	175.24	500.80	192.47
1880	1420	471.46	135.27	480.76	150.29	489.75	166.38	498.46	183.55	506.89	201.8
1900	1440	476.95	141.13	486.41	157.00	495.56	174.00	504.42	192.17	512.99	211.5

(Continued)

Table L.2 British Engineering (BE) Units *Continued*

<i>T</i>		<i>f</i> = 0		<i>f</i> = 0.0169		<i>f</i> = 0.0338		<i>f</i> = 0.0507		<i>f</i> = 0.0676	
°R	°F	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>
1920	1460	482.46	147.19	492.08	163.93	501.38	181.90	510.39	201.1	519.11	221.6
1940	1480	487.97	153.46	497.75	171.11	507.21	191.09	516.37	210.4	525.23	232.1
1960	1500	493.50	159.93	503.44	178.55	513.05	198.57	522.36	220.0	531.37	243.0
1980	1520	499.03	166.62	509.13	186.23	518.91	207.4	528.37	230.0	537.53	254.3
2000	1540	504.57	173.53	514.84	194.2	524.77	216.4	534.38	240.4	543.69	266.0
2020	1560	510.12	180.66	520.55	202.4	530.64	225.9	540.41	251.1	549.87	278.2
2040	1580	515.68	188.02	526.28	210.9	536.53	235.6	546.45	262.2	556.06	290.8
2060	1600	521.24	195.62	532.01	219.7	542.42	245.7	552.50	273.8	562.26	303.9
2080	1620	526.82	203.5	537.75	228.7	548.33	256.1	558.56	285.7	568.47	317.5
2100	1640	532.40	211.6	543.50	238.1	554.24	266.9	564.63	298.0	574.70	331.5
2120	1660	537.99	219.9	549.26	247.8	560.16	278.1	570.72	310.8	580.93	346.1
2140	1680	543.59	228.5	555.03	257.8	566.10	289.6	576.81	324.0	587.18	361.2
2160	1700	549.20	237.4	560.81	268.1	572.04	301.5	582.91	337.7	593.44	376.8
2180	1720	554.82	246.5	566.60	278.7	578.00	313.8	589.03	351.9	599.71	393.0
2200	1740	560.44	255.9	572.39	289.7	583.96	326.5	595.15	366.5	605.99	409.8
2220	1760	566.07	265.6	578.20	301.0	589.93	339.6	601.28	381.6	612.28	427.1
2240	1780	571.71	275.6	584.01	312.6	595.91	353.1	607.43	397.2	618.58	445.1
2260	1800	577.35	285.8	589.83	324.6	601.90	367.1	613.58	413.4	624.89	463.6
2280	1820	583.00	296.4	595.66	337.0	607.90	381.5	619.74	430.0	631.21	482.8

2300	1840	588.66	307.3	601.49	349.8	613.90	396.4	625.91	447.3	637.54	502.6
2320	1860	594.33	318.5	607.34	362.9	619.92	411.7	632.10	465.0	643.89	523.1
2340	1880	600.00	330.0	613.19	376.5	625.94	427.5	638.29	483.4	650.24	544.3
2360	1900	605.68	341.8	619.05	390.4	631.97	443.8	644.49	502.3	656.60	566.2
2380	1920	611.37	354.0	624.91	404.7	638.01	460.6	650.69	521.8	662.97	588.8
2400	1940	617.06	366.5	630.78	419.5	644.06	477.9	656.91	542.0	669.35	612.1
2420	1960	622.76	379.4	636.66	434.7	650.12	495.7	663.14	562.8	675.74	636.2
2440	1980	628.46	392.6	642.55	450.3	656.18	514.1	669.37	584.2	682.14	661.1
2460	2000	634.17	406.2	648.45	466.4	662.25	533.0	675.61	606.3	688.55	686.7
2480	2020	639.89	420.2	654.35	482.9	668.33	552.4	681.86	629.1	694.97	713.2
2500	2040	645.62	434.5	660.26	499.9	674.42	572.5	688.12	652.5	701.39	740.5
2520	2060	651.34	449.2	666.17	517.4	680.51	593.1	694.39	676.7	707.83	768.6
2540	2080	657.08	464.3	672.09	535.4	686.61	614.3	700.66	711.6	714.27	797.7
2560	2100	662.82	479.8	678.02	553.8	692.72	636.1	706.95	727.2	720.72	827.6
2580	2120	668.57	495.7	683.95	572.8	698.83	658.6	713.24	753.6	727.18	858.4
2600	2140	674.32	512.0	689.89	592.3	704.96	681.7	719.53	780.8	733.65	890.2
2620	2160	680.08	528.8	695.84	612.3	711.09	705.4	725.84	808.7	740.13	923.0
2640	2180	685.84	546.0	701.79	632.8	717.22	729.8	732.15	837.5	746.61	956.7
2660	2200	691.61	563.6	707.75	653.9	723.36	754.9	738.47	867.1	753.11	991.4
2680	2220	697.38	581.6	713.72	675.6	729.51	780.6	744.80	897.6	759.61	1027.2
2700	2240	703.16	600.2	719.69	697.8	735.67	807.1	751.14	928.9	766.12	1064.0

(Continued)

Table L.2 British Engineering (BE) Units *Continued*

<i>T</i>		<i>f</i> = 0		<i>f</i> = 0.0169		<i>f</i> = 0.0338		<i>f</i> = 0.0507		<i>f</i> = 0.0676	
°R	°F	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>
2720	2260	708.95	619.2	725.66	720.6	741.83	834.3	757.48	961.1	772.63	1101.9
2740	2280	714.74	638.6	731.65	744.0	748.00	862.2	763.83	994.2	779.16	1141.0
2760	2300	720.53	658.6	737.63	768.0	754.18	890.9	770.19	1028.3	785.69	1181.1
2780	2320	726.33	679.0	743.63	792.6	760.36	920.4	776.55	1063.3	792.23	1222.4
2800	2340	732.14	699.9	749.63	817.9	766.55	950.6	782.92	1099.2	798.77	1264.9
2820	2360	737.95	721.3	755.63	843.8	772.74	981.6	789.30	1136.1	805.33	1308.5
2840	2380	743.77	743.3	761.65	870.3	778.94	1013.5	795.68	1174.1	811.89	1353.4
2860	2400	749.59	765.8	767.66	897.5	785.15	1046.2	802.07	1213.1	818.46	1399.6
2880	2420	755.41	788.8	773.68	925.4	791.36	1079.7	808.47	1253.1	825.03	1447.1
2900	2440	761.24	812.3	779.71	954.0	797.58	1114.1	814.87	1294.2	831.61	1495.9
2920	2460	767.08	836.4	785.74	983.2	803.80	1149.3	821.28	1336.4	838.20	1546.0
2940	2480	772.91	861.1	791.78	1013.2	810.03	1185.5	827.69	1379.7	844.80	1597.5
2960	2500	778.76	886.4	797.82	1044.0	816.27	1222.6	834.12	1424.1	851.40	1650.4
2980	2520	784.61	912.2	803.87	1075.4	822.51	1260.6	840.54	1469.7	858.01	1704.7
3000	2540	790.46	938.6	809.93	1107.6	828.75	1299.6	846.98	1516.5	864.63	1760.5
3020	2560	796.32	965.7	815.98	1140.6	835.01	1339.5	853.42	1564.5	871.25	1817.9
3040	2580	802.18	993.3	822.05	1174.4	841.27	1380.5	859.87	1613.8	877.88	1876.7
3060	2600	808.05	1021.6	828.12	1209.0	847.53	1422.4	866.32	1664.3	884.51	1937.0
3080	2620	813.92	1050.5	834.19	1244.3	853.80	1465.4	872.78	1716.1	891.15	1999.0

3100	2640	819.79	1080.0	840.27	1280.6	860.07	1509.4	879.24	1769.2	897.80	2063
3120	2660	825.67	1110.2	846.35	1317.6	866.35	1554.4	885.71	1823.6	904.45	2128
3140	2680	831.55	1141.1	852.44	1355.5	872.63	1600.6	892.18	1879.4	911.11	2195
3160	2700	837.44	1172.6	858.53	1394.3	878.92	1647.8	898.66	1936.5	917.78	2263
3180	2720	843.33	1204.9	864.62	1433.9	885.22	1696.2	905.15	1995.1	924.45	2334
3200	2740	849.23	1237.8	870.73	1474.5	891.52	1745.8	911.64	2055	931.13	2406
3220	2760	855.13	1271.5	876.83	1515.9	897.82	1796.4	918.14	2117	937.81	2480
3240	2780	861.03	1305.8	882.94	1558.3	904.13	1848.3	924.64	2180	944.50	2556
3260	2800	866.94	1340.9	889.06	1601.7	910.45	1901.4	931.15	2244	951.20	2634
3280	2820	872.85	1376.8	895.17	1646.0	916.77	1955.7	937.66	2310	957.90	2714
3300	2840	878.77	1413.4	901.30	1691.2	923.09	2011	944.18	2378	964.60	2795
3320	2860	884.69	1450.7	907.43	1737.5	929.42	2068	950.70	2447	971.31	2879
3340	2880	890.61	1488.8	913.56	1784.8	935.75	2126	957.23	2518	978.03	2965
3360	2900	896.54	1527.8	919.69	1833.1	942.09	2186	963.76	2590	984.75	3053
3380	2920	902.47	1567.5	925.83	1882.4	948.43	2246	970.30	2665	991.48	3143
3400	2940	908.40	1608.1	931.98	1932.8	954.78	2309	976.84	2741	998.21	3235
3420	2960	914.34	1649.4	938.12	1984.3	961.13	2372	983.39	2818	1004.95	3329
3440	2980	920.28	1691.6	944.27	2037	967.48	2437	989.94	2898	1011.69	3426
3460	3000	926.23	1734.7	950.43	2091	973.84	2503	996.49	2979	1018.43	3524
3480	3020	932.18	1778.6	956.59	2145	980.20	2571	1003.05	3062	1025.18	3626
3500	3040	938.13	1823.4	962.75	2201	986.57	2640	1009.62	3147	1031.94	3730

(Continued)

Table L.2 British Engineering (BE) Units *Continued*

<i>T</i>		<i>f</i> = 0		<i>f</i> = 0.0169		<i>f</i> = 0.0338		<i>f</i> = 0.0507		<i>f</i> = 0.0676	
°R	°F	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>	<i>h_r</i> Btu/lbm	<i>P_r</i>
3520	3060	944.08	1869.1	968.92	2258	992.94	2711	1016.19	3234	1038.70	3836
3540	3080	950.04	1915.7	975.08	2317	999.31	2783	1022.76	3323	1045.46	3944
3560	3100	956.00	1963.2	981.26	2376	1005.69	2857	1029.34	3414	1052.23	4055
3580	3120	961.96	2012	987.43	2437	1012.07	2933	1035.92	3507	1059.01	4169
3600	3140	967.93	2062	993.61	2499	1018.46	3010	1042.50	3602	1065.78	4285
3620	3160	973.90	2111	999.80	2562	1024.85	3088	1049.09	3699	1072.57	4404
3640	3180	979.87	2163	1005.98	2627	1031.24	3169	1055.68	3798	1079.35	4525
3660	3200	985.85	2215	1012.17	2692	1037.63	3251	1062.28	3899	1086.14	4649
3680	3220	991.83	2268	1018.36	2759	1044.03	3334	1068.88	4003	1092.93	4776
3700	3240	997.81	2322	1024.56	2828	1050.44	3420	1075.48	4109	1099.73	4906
3720	3260	1003.79	2378	1030.76	2897	1056.84	3507	1082.09	4217	1106.53	5039
3740	3280	1009.78	2434	1036.96	2969	1063.25	3596	1088.70	4327	1113.34	5174
3760	3300	1015.77	2492	1043.16	3041	1069.66	3686	1095.31	4439	1120.15	5313
3780	3320	1021.76	2550	1049.37	3115	1076.08	3779	1101.93	4554	1126.96	5455

3800	3340	1027.76	2610	1055.58	3190	1082.50	3873	1108.55	4672	1133.77	5599
3820	3360	1033.75	2670	1061.80	3267	1088.92	3970	1115.17	4792	1140.59	5747
3840	3380	1039.75	2732	1068.01	3345	1095.34	4068	1121.80	4914	1147.41	5898
3860	3400	1045.76	2795	1074.23	3425	1101.77	4168	1128.43	5039	1154.24	6052
3880	3420	1051.76	2859	1080.45	3506	1108.21	4270	1135.06	5166	1161.07	6210
3900	3440	1057.77	2924	1086.68	3589	1114.64	4374	1141.70	5296	1167.91	6371
3920	3460	1063.78	2990	1092.91	3673	1121.08	4481	1148.34	5429	1174.74	6535
3940	3480	1069.80	3058	1099.14	3759	1127.52	4589	1154.99	5564	1181.58	6703
3960	3500	1075.81	3126	1105.37	3847	1133.96	4699	1161.63	5702	1188.43	6874
3980	3520	1081.83	3196	1111.61	3936	1140.41	4812	1168.29	5843	1195.28	7049
4000	3540	1087.86	3268	1117.85	4027	1146.86	4927	1174.94	5987	1202.13	7227

